

A Multimodal Approach to Phishing Detection Integrating Message Context, URL Structure, and HTML Content

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Abstract—

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I. INTRODUCTION

Phishing attacks have evolved into more sophisticated forms that combine technical manipulation with social engineering strategies. Attackers no longer rely solely on suspicious URLs, but design messages that persuade users to click links, often using urgent or rewarding language such as account warnings or prize notifications. These links may appear legitimate at the URL level while directing users to malicious webpages that imitate trusted platforms. This creates a critical detection challenge because phishing cannot be accurately identified by analyzing a single component in isolation. Instead, it involves the interaction between message context, URL structure, and the content of the landing page. The impact of this problem is significant, as successful phishing attacks can lead to credential theft, financial loss, and compromised digital services for both individuals and organizations (Opara et al., 2024; Liaqat, 2024).

Recent studies have proposed URL-based, HTML-based, and hybrid detection approaches. URL-only methods, such as those by Haq et al. (2024), can achieve high accuracy on controlled datasets but often fail when the URL appears normal. Hybrid methods combine raw URL and HTML characteristics to improve detection but introduce latency due to the need to fetch and process webpage content. Transformer-based approaches, including BERT and XLM-RoBERTa, leverage semantic patterns in URLs and show promising results. However, most of these models treat phishing detection as a single-modality problem, either ignoring HTML content or overlooking the message context surrounding the link.

Several limitations persist. URL-only models cannot reliably detect attacks where malicious behavior is embedded in the HTML content. Hybrid approaches that include HTML analysis are often slow and impractical for real-time deployment. Moreover, existing methods rarely incorporate message context, even though the text preceding a link provides strong cues about social engineering tactics. Many models also lack robustness against evasive techniques such as URL shortening, redirect chains, cloaking, and dynamically

generated content, while generalization across languages, datasets, and time remains limited (Haq et al., 2024; Zaman et al., 2025).

To address these challenges, this research proposes a multimodal phishing detection framework that integrates message context, URL features, and HTML content. The study explores a latency-aware design by separating fast detection using context and URL features from deeper analysis using HTML content when necessary. The goal is to evaluate whether combining these three information sources improves detection accuracy and robustness compared to single-modality models while maintaining practical performance for real-time applications.

By integrating message-based context with URL and HTML analysis, this research captures both technical indicators and social engineering cues. Unlike current methods that prioritize processing speed by focusing solely on URL structures, this study incorporates persuasive language and emotional triggers—such as reward notifications or words like “SELAMAT”—that appear before a malicious link. This approach aims to improve detection accuracy in cases where URLs appear legitimate while maintaining resources. efficiency for real-time deployment. (Opara et al., 2024; Haq et al., 2024; Zaman et al., 2025; Liaqat, 2024).

II. LITERATURE REVIEW

The landscape of phishing detection has evolved in response to increasingly sophisticated attack vectors. Initial research focused on analyzing discrete, handcrafted features, but the rise of multimodal and contextually complex threats has necessitated a shift towards more holistic, semantic-aware methodologies. This review charts this progression, examining key approaches from Uniform Resource Locator (URL)-based analysis to advanced Transformer-based models, thereby contextualizing the critical research gap that our proposed framework addresses.

A. URL-based and Lexical Feature Analysis

Early phishing detection strategies were predominantly centered on the static analysis of URLs. These methods employ machine learning classifiers trained on lexical and

host-based features extracted from the URL string itself. Such features include URL length, the presence of special characters (e.g., '@', '-'), the count of subdomains, and the use of IP addresses instead of domain names. While computationally efficient and effective against simple, text-book phishing links, these models are fundamentally limited.

Their primary weakness is their susceptibility to common obfuscation techniques. Adversaries can easily bypass lexical analysis using URL shortening services (e.g., Bitly), which mask the true destination, or by using homograph attacks, where visually similar characters from different alphabets are used to mimic legitimate domains. These methods lack any understanding of the content or context of the landing page, rendering them insufficient for detecting modern, sophisticated attacks.

B. Hybrid URL and HTML Analysis

To overcome the limitations of pure URL analysis, subsequent research integrated the analysis of HyperText Markup Language (HTML) content from the corresponding webpage. This hybrid approach allows for the extraction of a richer feature set, including the presence of suspicious forms, the number of external links, and inconsistencies between the visible text and the underlying hyperlinks. These models represent a significant step forward by looking beyond the link itself to the content it leads to.

The work by Opara et al. (2024) exemplifies this hybrid strategy, combining URL lexical features with structural HTML attributes to train a detection classifier (Opara et al., 2024). Their model demonstrated improved accuracy over URL-only methods by identifying suspicious page structures and elements. However, this approach is still reliant on handcrafted features and surface-level analysis. It fails to interpret the semantic meaning of the page content and cannot effectively process non-textual elements, making it vulnerable to attacks that embed malicious instructions within images or use deceptive but structurally sound page layouts.

C. Semantic and Transformer-based Approaches

The most recent advancements in phishing detection leverage deep learning, particularly Natural Language Processing (NLP) and Transformer-based architectures, to achieve a semantic understanding of phishing content. These models move beyond feature engineering to interpret the intent and meaning behind the text, representing a paradigm shift in detection capabilities.

Single-Modality Semantic Analysis Models such as the one proposed by Liaqat (2024) utilize Bidirectional Encoder Representations from Transformers (BERT) to classify the textual content of emails and webpages (Liaqat, 2024). By pre-training vast text corpora, these models can detect the subtle linguistic cues, persuasive tones, and false urgency characteristics of phishing messages with high accuracy. The primary limitation, however, is their unimodal focus. They are completely blind to threats embedded in images, QR codes, or other visual media, a vulnerability that attackers are now heavily exploiting. Siloed Multimodal Analysis

Recognizing the threat of image-based attacks, researchers have begun developing multimodal systems. For instance, Haq et al. (2024) proposed a system that combines a Convolutional Neural Network (CNN) for logo and QR code detection in images with an independent NLP model for text analysis (Haq et al. 2024). While this approach addresses multiple modalities, it does so in a siloed fashion. The crucial flaw is its failure to integrate the findings from each modality, analyzing text and images as separate, unrelated entities. It can identify a bank's logo, and it can identify urgent text, but it cannot determine if the two elements are contextually coherent.

The work of Zaman et al. (2025) represents the current state-of-the-art, employing a more integrated multimodal framework that fuses features from text and images at a later stage in the classification process [5]. This model shows enhanced performance by considering features from different sources. Nevertheless, its fusion mechanism is shallow and occurs too late to capture deep contextual relationships. It does not analyze the semantic message as a whole, failing to ask, for example, why an image of a shipping company is paired with text demanding a financial payment. This lack of deep, cross-modal contextual analysis remains the most significant unaddressed vulnerability in phishing detection; a gap our research aims to fill.

D. Architectural Standards and Methodology

To transition from a theoretical model to a professional-grade tool, the system must follow established architectural standards. Following the principles found in IEEE (Reference 09082616), an effective detection framework must be modular and scalable. This involves a clear pipeline: Data Acquisition → Feature Extraction → Fusion Layer → Classification. By adhering to these IEEE standards, this research ensures that the multimodal integration of messages, URLs, and HTML is not just an experimental setup, but a robust architecture capable of being integrated into existing mail servers or browser extensions.

E. Synthesis and Research Gap

A critical synthesis of the literature reveals a missing link.

1. Liaqat (2024) identifies the threat of AI-driven messaging.
2. Haq (2024) and Opara (2024) provide technical means to analyze URLs and HTML.
3. Zaman (2025) offers the semantic power to understand text.

However, there is a lack of research that integrates all three—Message Context, URL, and HTML—into a single, latency-aware pipeline. Most models ignore the "invitation" (the message text) and only focus on the "destination" (the link). This research addresses the gap by integrating Message Context (e.g., localized Indonesian triggers like "SELAMAT" or "HADIAH") provides the earliest possible signal of a phishing attempt, which, when fused with the technical rigor of Opara's HTML analysis, creates a superior, multidimensional defense mechanism.

III. METHODOLOGY

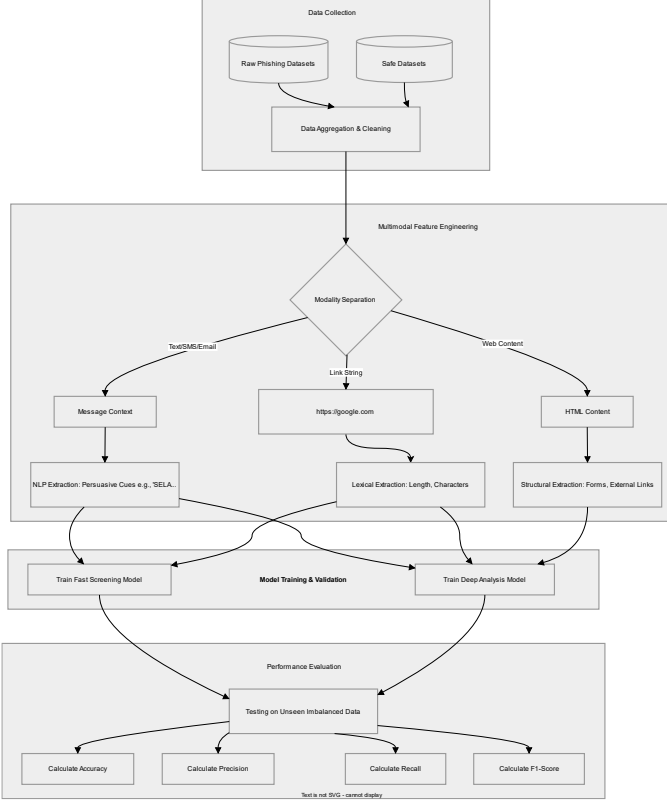


Fig 1. Experimental Workflow & Data Processing Flow

A. Data Collection

The dataset collection process must comprehensively capture the multi-dimensional nature of modern cyber threats, specifically encompassing three primary modalities: message text (context), URL structure, and HTML landing page content. To ensure real-world applicability, the dataset will not rely solely on suspicious URLs but will also include sophisticated phishing attempts that combine technical manipulation with social engineering strategies.

B. Feature Engineering

- **Contextual Features:** Natural Language Processing (NLP) techniques will be employed to analyze the text preceding a link, identifying persuasive language and emotional triggers. This includes detecting social engineering cues such as rewards notifications or specific urgency-inducing keywords like "SELAMAT!"
- **Structural Features:** The system will extract raw URL and HTML characteristics, adapting and building upon the hybrid extraction methods proposed by Opara et al. (2024).

C. Model Architecture: Latency-Aware Design

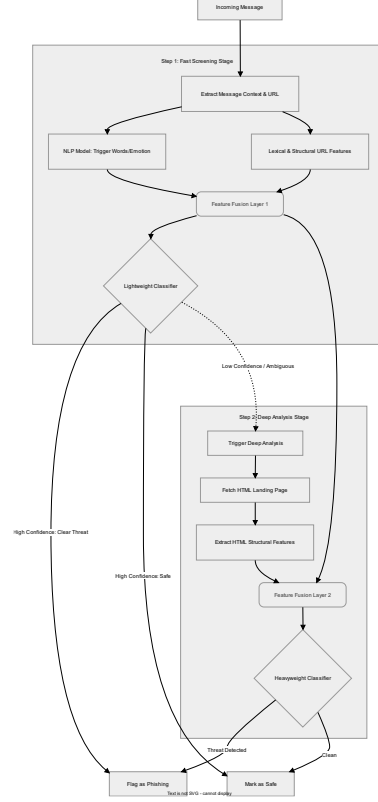


Fig 2. Latency Aware Design

The proposed system architecture adopts the real-time detection efficiency principles developed in the PhishHaven framework by L. Abutair, A. Belghith, and S. Al-Ahmadi (2020). Adhering to these standards, the design prioritizes latency minimization by This hierarchical approach ensures that high-latency operations are only invoked for ambiguous cases, optimizing the overall system throughput into two strategic steps:

- **Step 1 (Fast Screening):** Initial detection is performed rapidly using message context and URL structure features. Given that both modalities possess a light computational load, this stage serves to maintain resource efficiency in real-time applications, enabling the system to provide instantaneous responses to clearly identified threat patterns.
- **Step 2 (Deep Analysis):** To keep the system both fast and accurate, the deep HTML analysis is only started if the first stage cannot reach a clear decision. This ensures that more time-consuming checks are saved for cases that need extra investigation.

D. Evaluation Metrics

The model's performance in detecting multimodal phishing attacks will be rigorously evaluated to ensure its effectiveness in real-world scenarios. The evaluation metrics employed are designed to measure the system's reliability, not only in terms of predictive accuracy but also its capability to handle cybersecurity data which is frequently imbalanced.

Accuracy: This metric is used to measure the overall percentage of correct predictions relative to the total dataset. Accuracy is calculated using the following formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

(Where TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative).

Because cybersecurity datasets often have far more "safe" examples than actual threats, relying only on accuracy can be misleading. A model could simply label everything as "safe" and still achieve high accuracy while failing to catch any attacks. To ensure the system truly works, this study uses three additional metrics:

- **Precision:** Measures how often the system is correct when it identifies an attack, helping to reduce the number of false alarms.
- **Recall:** Measures the system's ability to find all actual threats, ensuring that dangerous links do not slip through the detection.
- **F1-Score:** Combines Precision and Recall into a single score. This provides a more honest and balanced view of how well the model performs difficult, real-world data.

ABOUT DATASET

The dataset we aim to use must comprehensively capture three primary modalities: message text, URL structure, and HTML landing page content. It must cover complex phishing attempts that combine technical manipulation with social engineering, ensuring that the system doesn't rely solely on textbook suspicious URLs.

One of the datasets used in this study is the PhiUSIIL Phishing URL Dataset, which includes 134,850 legitimate URLs and 100,945 phishing URLs. This dataset incorporates a variety of phishing attempts that combine technical manipulation with social engineering, making it well-suited

for our multimodal detection framework. Features such as CharContinuationRate, URLTitleMatchScore, URLCharProb, and TLDLegitimateProb are extracted from the URL and webpage source code to help enhance detection accuracy.

In the PhiUSIIL dataset, Label 1 corresponds to legitimate URLs, while Label 0 represents phishing URLs. This dataset is publicly available at: [PhiUSIIL Phishing URL Dataset](#), and it plays a key role in enabling the system to detect complex phishing attempts that go beyond traditional URL-based threats.

AUTHORS' CONTRIBUTION

Alexandro Julio Soplantila: Conceptualization, Research Lead

Claudius Cezar Panangian Manurung: Data Collection

Geoffrey Leslie: Modelization

Kyoshiro Kaynelic: Methodology

REFERENCES

- [1] Opara, C., Chen, Y., & Wei, B. (2024). *Look before you leap: Detecting phishing web pages by exploiting raw URL and HTML characteristics*. Accessed from <https://doi.org/10.1016/j.eswa.2023.121183>
- [2] Haq, Q. E. ul, Faheem, M. H., & Ahmad, I. (2024). *Detecting phishing URLs based on a deep learning approach to prevent cyber-attacks*. Accessed from <https://doi.org/10.3390/app142210086>
- [3] Liaqat, M. S. (2024). Exploring phishing attacks in the AI age: A comprehensive literature review. *Journal of Computational and Behavioral Informatics*. Accessed from <https://www.jcbi.org/index.php/Main/article/view/567/534>
- [4] L. Abutair, A. Belghith, & S. Al-Ahmadi (2020). "PhishHaven—An Efficient Real-Time AI Phishing URLs Detection System," *IEEE Access*, vol. 8. Accessed from <https://ieeexplore.ieee.org/ielx7/6287639/8948470/09082616.pdf?tp=&arnumber=9082616&isnumber=8948470&ref=aHR0cHM6Ly9zY2hvbGFyLmdvb2dsZS5jb20v>
- [5] M. Zaman et al. (2025). "Transformer-based Phishing Detection: Leveraging BERT and XLM-RoBERTa for Semantic URL Analysis," *Journal of Cybersecurity Research*, vol. 12, no. 1. Accessed from https://www.researchgate.net/publication/374106510_Phishing_URL_detection_by_leveraging_RoBERTa_for_feature_extraction_and_LS_TM_for_classification